

- 18** High concentrations of ethanol disrupt cell membrane structure by denaturing proteins and increasing the separation of adjacent phospholipid molecules. As a result, cell membranes can decrease in thickness by up to 30% and become more permeable.

Yeast cells release ethanol as a waste product of metabolism. In response to increased ethanol concentration in their environment, yeast cells are able to increase the tolerance of their cell membranes to ethanol.

Which statement correctly explains a response to ethanol that could account for the increase in tolerance of yeast cell membranes to ethanol?

- A** Decreasing the ratio of saturated fatty acids to unsaturated fatty acids within cell membranes helps to prevent the tails of phospholipids on one side of the bilayer from sliding past the tails of phospholipids on the other side of the bilayer.
- B** Increasing the proportion of palmitoleic acid (a C16 unsaturated fatty acid) to oleic acid (a C18 unsaturated fatty acid) in the phospholipids of the bilayer increases the fluidity of the cell membrane.
- C** Activating a cell-signalling pathway triggers the unfolded protein response in the endoplasmic reticulum. The unfolded protein response pauses protein synthesis and initiates cell death in yeast cells with a high proportion of mis-folded proteins.
- D** Increasing the proportion of ergosterol in the cell membrane prevents the accumulation of polar molecules, such as ethanol, within the cell. Ergosterol in yeast cells has a similar effect on membrane permeability as cholesterol in mammalian cells.